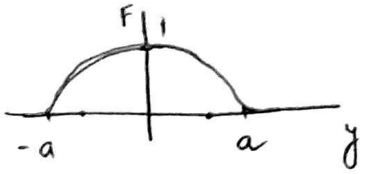
Prob 4 (Z&T. Ch 8. 1.2)

Draw sol-ns to the 1-dim-l wave eq-n of the form

(1) $u(x,t) = G(x+t)$ where

(2) $G(y) = F(y)$ is the "blip" f-n :

(3) 

$\longrightarrow$ "blip" f-n.

(i.e $F(y)$ is 1 when $y=0$ & goes to zero so that $F(y) \to 0$ for $y \le -a$ & $y \ge a$).

Draw the sol-n at $t = 0, 1, 2, \& 3$. Then sketch the region in the xt plane where the sol-n is <u>not</u> zero.

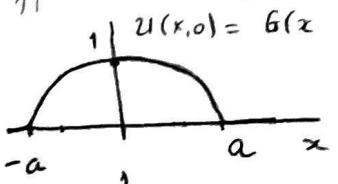
Soln:

[Preamble]

- Let's go one step at a time with different t values:

(i) $t = 0$:

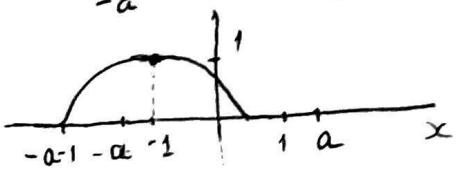
$u(x,0) = G(x+0) = G(x)$



$\Leftarrow \boxed{t=0}$

(ii) $t = 1$

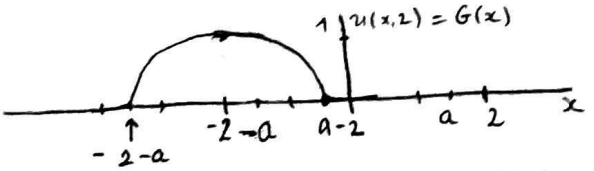
$u(x,1) = G(x+1)$



$\Leftarrow \boxed{t=1}$

(iii) $t = 2$

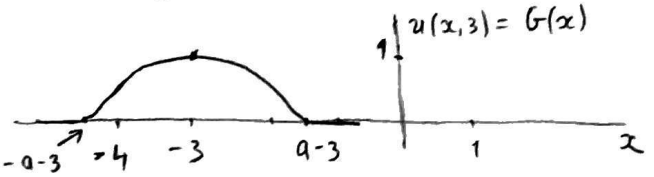
$u(x,2) = G(x+2)$



$\Leftarrow \boxed{t=2}$

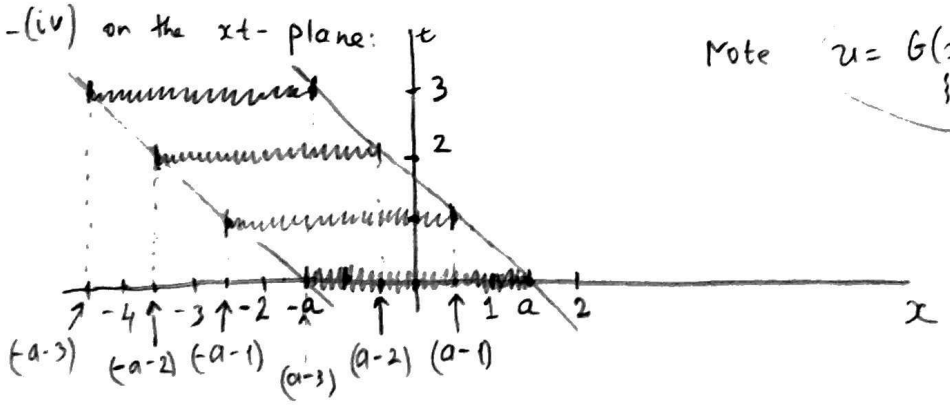
(iv) $t = 3$

$u(x,3) = G(x+3)$



$\Leftarrow \boxed{t=3}$

- Sketching (i)-(iv) on the xt- plane:



Note $u = G(x+t)$
$\S_x \; \S_t$

?2